

GTU CSE 241 / 501 Quiz

Std Code:210

Std Number:

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```
using namespace std;
```

```
class Person {
```

```
    public :
```

```
    Person() : name( "" ), age( 0 ) { } //default
```

```
    Person(const string& nam, int data ) : name(nam) , age( data) { if ( age < 0 ) exit( 1 );
```

```
    void set_name( const string& data ) { name = data ; }
```

```
    string get_name( ) const { return name; }
```

```
    void set_age(int data ) { age = data ; }
```

```
    int get_age( ) const { return age ; }
```

```
    Person(const Person& p ) : name(p.name) , age(p.age) { } // copy constructor;
```

```
    friend ostream& operator<<(ostream& os , const Person& p ) { os << p.name << " , " <<
```

```
    p.age << endl;
```

```
    return os; }
```

```
    protected : // to access in derived class
```

```
        string name;
```

```
        int age ; //checked for negative numbers
```

```
};
```

```

class Student : public Person {

    public :

        Student( ) : Person( ) , gpa( 0.0 ) { }

        Student ( const string& nam , int ag , double gp ) : Person(nam , ag ) , gpa( gp ) {
            if ( gpa < 0.0 ) exit( 1 ) ;
        }

        Student(double g ) : Person( ) , gpa( g ) {

            if( g < 0 ) exit( 1 ) ; }
        Student( const Student & st ) : Person(st ) , gpa( st.gpa) { }

        void set_gpa( double gp ) {
            if( gp < 0 . 0 ) exit( 1 ) ;
            gpa = gp;
        }

        double get_gpa( ) const { return gpa ; }

        friend ostream& operator << ostream& os , const Student& st ) {
            os << get_name ( ) << “ is “ << get_age( ) << “ years old and he has “ << gpa << endl;
            return os ; {
//WE COULD DIRECTLY USED NAME AND AGE .

        private :

            double gpa //I checked invalid gpa.
};

```

